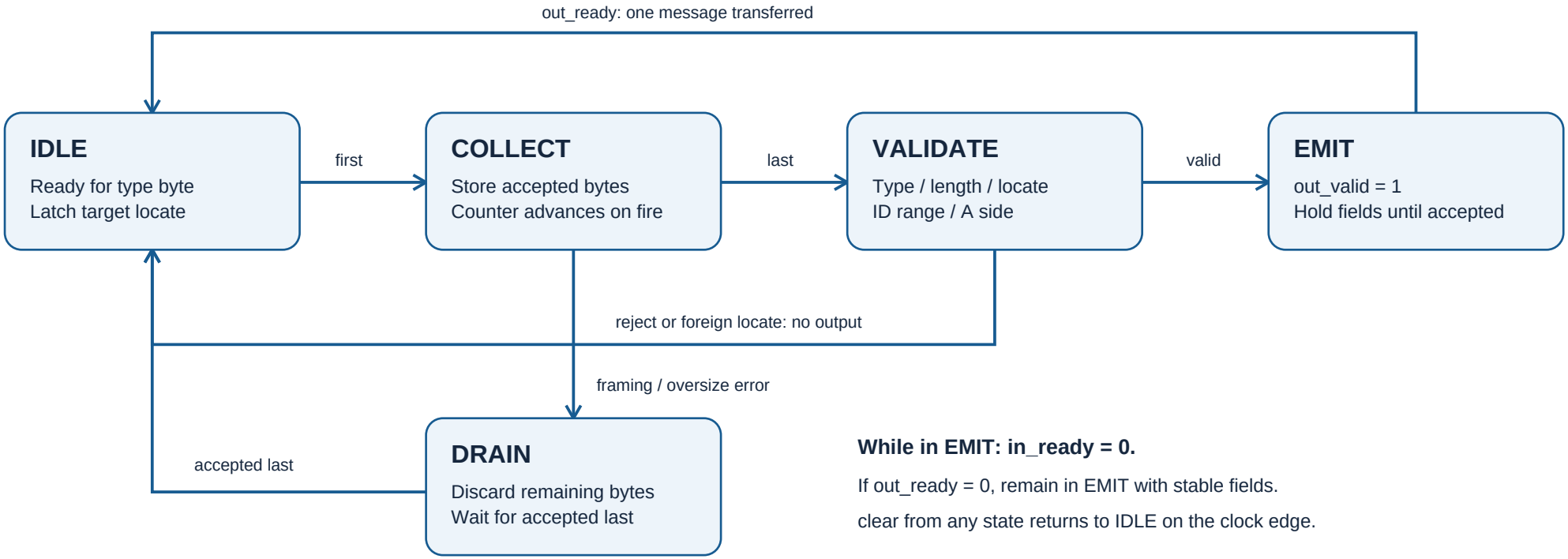
ITCH parser - state-machine sketch

Amirlan / CompE | November 8, 2026 | Implemented decoder

Input: one framed ITCH body. A byte is accepted only when in_valid AND in_ready.



While in EMIT: in_ready = 0.

If out_ready = 0, remain in EMIT with stable fields.

clear from any state returns to IDLE on the clock edge.

Boundary cases

First and last together: IDLE -> VALIDATE. Missing first: IDLE -> DRAIN (or IDLE if also last).

In COLLECT, a second first or a 37th byte rejects the frame. If that byte is also last, return directly to IDLE.

Unsupported type, wrong length, bad side or out-of-range ID never emits an update. Foreign locate is filtered.

Worked Add Order

36 accepted bytes: A / locate 1 / ID 42 / Buy / 100 shares / price 1,502,500 ($150.2500).

After last byte at edge L: validate at L+1; earliest output transfer at L+2. Gaps and stalls add time.

Scope: hw/itch_parser.ml. MoldUDP64 framing is a separate proposed adapter, not implemented here.

Full contract and transport FSM: docs/parser-interface.md. Tests: hw/test/test_itch_parser.ml.